Google Cloud

Cloud Data Analytics glossary

Terms and definitions from all courses in the certificate

A

Abstraction: A concept used to understand complex concepts and ideas by focusing on the most essential parts

Advanced provisioning: A cloud delivery model in which the user signs a formal agreement with the cloud service provider and either pays a set price, or is billed monthly

Alerting: A feature that enables dashboard users to receive a notification when predetermined conditions are met or exceeded

Algorithm: A process or set of rules followed for a specific task

Apache Hadoop: A framework that distributes the processing of data across clusters of computers

Application programming interface (API): A protocol that enables one application to interact with another application

Architecture: A framework that defines the design of a technical solution

Archival policy: An outline of where and how data is stored once an analysis project is complete

Array: A collection of values of the same data type

Audit: A formal examination of how users are accessing data in order to ensure safe and appropriate access, while identifying and solving any data concerns

Automated transformation: The use of processing and scripting tools with less or no programming, compared to manual transformation

Automatic refreshing: A process that enables dashboards to be automatically updated at regular intervals

Automation: The use of software, scripting, or machine learning to perform data analysis processes without human work

Autoscaling: A cloud service that monitors applications, automatically scaling up or down according to the computing resources needed to meet user demand

B

Backend platform: The components of cloud architecture that make up the cloud, including computing resources, storage, security mechanisms, and management

Batch ingestion: A method of collecting data over time, and then processing it in groups, also called batches.

Batch loading: A method where data is moved to destination storage in groups called batches at a predetermined schedule

Batch processing: A method of collecting large volumes of data over a period of time, then processing it all at once

BigQuery: A data warehouse on Google Cloud used to query data, filter large datasets, aggregate results, and perform complex operations

Boolean: A data type with only two possible values, usually true or false

Branch: A working copy of a repository

Business data request: A business question that can be answered with data

C

Caching: The process of storing data in a temporary location so that it can be accessed more quickly in the future

Caching policy: A set of rules that determine how long cached results are stored, and when they are refreshed

Categorical data (Qualitative data): A subjective and explanatory measure of a quality or characteristic

Check-in: Upload code to a main repository so others can access it and review it

Classification: The process of grouping items into a predefined category based on common characteristics

Cloud computing: The practice of using on-demand computing resources as services hosted over the internet

Cloud cost optimization: The process of reducing cloud expenses by implementing cost-reduction strategies

Cloud data analytics: The process of collecting, storing, managing, organizing, transforming, and analyzing large volumes of data using cloud-based services and solutions

Cloud data storage: A solution that enables organizations to keep, access, and maintain digital data in off-site, cloud-based servers

Cloud data warehouse: A large-scale, data storage solution hosted on remote servers by a cloud service provider

Clustered column: A user-defined table property that arranges storage blocks based on the values within their columns

Clustered table: A table in which column order is defined by the user, using clustered columns

Clustering: Grouping data points that are similar to each other in some way

Collaboration: Members of the data analytics team visit the central system to review data requests, including queries and data delivery methods

Comma-Separated Values (CSV) file: A delimited text file that uses a comma to separate field values

Context: The condition in which something exists or happens

Credentials: A username and password

D

Dashboards as code: An approach to managing dashboards by defining them in code

Data access management: The process of implementing features including password protection, user permissions, and encryption to protect data and related processes

Data aggregation: The process of gathering data and expressing it in a summary form

Data analyst: A professional who performs data analysis workflows, including importing, manipulating, calculating, and reporting on business data

Data analyst portfolio: A website that showcases who you are, your skills, and the projects you've worked on

Data architect: A professional who collaborates with data analysts, data scientists, and data engineers to design the infrastructure of a database

Data blending: The process of combining data from multiple data sources to create a single report visualization

Data catalog: A centralized inventory of an organization's data assets

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Data center: A physical building that contains servers, computer systems, and associated components

Data cleaning: The process of fixing or removing data quality issues

Data connection: A link formed between a data source and its storage location

Data conversion: The process of changing the format of data to improve compatibility, readability, or make data more secure

Data curation: The process of selecting and organizing data to ensure it’s useful and of high quality

Data derivation: The process of combining and processing existing data using an algorithm to create new data

Data discovery: The process of finding and understanding data assets within a dataset or data ecosystem, and identifying relevant patterns and relationships

Data documentation: A written guide to the data contained in a dataset, how it was collected, and how it's organized

Data drilling: A process that allows users to explore data in more detail by revealing additional levels of information

Data engineer: A professional who transforms data into a useful format for analysis, and gives it a reliable infrastructure

Data enrichment: The process of adding additional information to data

Data exploration: The process of understanding a dataset by inspecting its characteristics, identifying patterns, and asking questions

Data freshness: How current, or up-to-date, the data within a report or dashboard is

Data governance: A process for ensuring the formal management of a company’s data

Data ingestion: The process of obtaining, importing, and processing data for later use or storage

Data integration: The process of combining data from different sources into a single, usable data source

Data integrity: The accuracy, completeness, consistency, and trustworthiness of data throughout its life cycle

Data lake: A database system that stores large amounts of raw data in its original format until it's needed

Data lakehouse: A hybrid data architecture that combines the features of a data lake with those of a data warehouse

Data lifecycle: The sequence of stages that data experiences, which include plan, capture, manage, analyze, archive, and destroy

Data literacy: The ability to understand and use data

Data management: The process of establishing and communicating a clear plan for collecting, storing, and using data

Data mapping: The process of matching fields from one data source to another

Data mapping rules: A set of instructions that define how the fields will be matched

Data mart: A subject-oriented database that can be a subset of a larger data warehouse

Data model: A concept for organizing data elements and how they relate to one another

Data modeling: The process of designing, structuring, joining, and transforming data to prepare it for reporting

Data modeling language: A tool used to create and represent semantic data models

Data pipeline: A series of processes that transport data from different sources to a destination for storage and analysis

Data privacy: Preserving a data subject's information any time a data transaction occurs

Data processing: Raw data is cleaned, organized, and changed into a format for easy analysis

Data profiling: The process of exploring data to identify quality issues

Data report: A visualization of detailed business intelligence data for making business decisions

Data retention: The collection, storage, and management of data

Data scientist: A professional who works primarily in data research, identifying business questions, collecting data from multiple sources, organizing it, and finding answers

Data silo: A collection of data held by one group at a central location

Data source: The place where data was generated

Data standardization: The process of ensuring that all the data in a dataset is in a common format

Data storage: The amount of data kept in storage spaces called “buckets”

Data transformation: The process of taking raw data and converting it into a usable format

Data unification: The process of integrating data from diverse sources to create a consolidated view

Data validation: The process of checking and rechecking the quality of data so that it's complete, accurate, secure, and consistent

Data visualization: The graphical representation of data using charts, graphs, and other visual formats

Data warehouse: A database that consolidates data from multiple source systems for data consistency, accuracy, and efficient access

Database partitioning: The process of dividing data into separate data segments or partitions that can be managed and accessed separately

Database schema: A way of describing how a database is organized

Data-collection policy: An outline that creates rules around how data is collected, and what resources may be used in the process

Data-driven culture: A culture in which both technical and nontechnical workers collaborate, and feel empowered to make decisions with data

Deduplication: The process of eliminating a dataset's redundant data

Deletion policy: An outline of when and how data is permanently destroyed

Denormalized data: Stores repeated information in one or more tables

Derived table: A query whose results are used as if it were an actual table in the database

Dimensional hierarchy: A way to define the levels of detail in a dataset that a chart can display

Dimensions: Unique attributes that help you to describe data

Documentation: A process for preserving all of the details and conversations about each request in one place

Drill through: A technique that allows users to navigate to related visualizations

Drilling down: A technique that reveals additional levels of detail within a chart by moving down the hierarchy from more general, to more granular data

Drilling up: A technique that reveals fewer levels of detail within a chart by moving up the hierarchy from more granular, to more general data

Duplicate data: A record that repeats the information–in whole or in part–of another record

Dynamic provisioning: A cloud delivery model in which resources are adjusted based on the user's changing needs, and they are only charged for what they use

E

Encryption: The process of encoding information

Enterprise-grade data visualization tools: A type of data visualization software used by large data-driven organizations to explore, analyze, and share business analytics

Exact duplicate: A record where all the data repeats another record

Extract: The stage of retrieving data from one or more sources in a data pipeline

Extract-load-transform (ELT): A type of data pipeline that enables data to be gathered from data lakes, loaded into a unified destination system, and transformed into a useful format

Extract-transform-load (ETL): A type of data pipeline that enables data to be gathered from source systems, converted into a useful format, and brought into a data warehouse, or other unified destination system

F

Filtering: A tool for limiting the data returned from a query based on specific criteria

Float: A number that contains a decimal

Foreign key: A column within a table that is a primary key in another table

Frontend platform: The component of cloud architecture that users interact with

Functional partitioning: A process for dividing data into separate segments, in which data is grouped based on how it's used in different parts of the database system

G

Google Colab: A cloud-hosted version of Jupyter Notebook that allows users to write and execute Python in a browser, with no configuration required

Guided analytics: An approach to business intelligence where solutions, like reports and dashboards, are created by the data team to meet a specific need

H

Hadoop: A framework that distributes the processing of data across clusters of computers

Historical records: Past data or information that's been stored and can be accessed or retrieved when needed

Hold: A policy placed on a dataset that prevents its deletion, or prevents deletion capabilities for certain accounts

Horizontal partitioning: A process for dividing data into separate segments, in which each partition or shard is its own data storage, but all shards follow the same organizational layout

Hybrid cloud: A cloud model that combines public and private models, so organizations can enjoy both cloud services, and the control features of on-premises cloud models

Hyperparameters: Parameters whose value is used to control the learning process

I

Identity Access Management (IAM): The process of assigning certain individual roles or groups with access to particular resources

Incremental loading: A method where only the data that has changed since the last load is moved to destination storage

Infrastructure as a service (IaaS): A cloud computing model that offers on-demand access to IT infrastructure services, including hardware, storage, networking, and virtualization tools

Inner join: A join that only returns the rows that match in both tables

Input data: Any information that a user sends to a computer

Integer: A data type that contains whole numbers

Integrated Development Environment (IDE): An application that brings together the tools needed for development in a user-friendly environment

J

JavaScript Object Notation (JSON): A text-based data-interchange format

L

Linear regression: A technique that estimates the linear relationship between a continuous dependent variable, and one or more independent variables

Live dashboard: A data visualization tool that provides near real-time data updates

Load: The stage of inserting data into the target database, data store, data warehouse, or data lake in a data pipeline

M

Machine learning: The use and development of algorithms and statistical models to teach computer systems to analyze and discover patterns in data

Managed service: The act of a third-party provider taking care of the ongoing maintenance, management, and support of an organization's cloud infrastructure and applications

Managed Service for Apache Spark: A fully managed service that maximizes open-source data tools for batch processing, querying, streaming, and machine learning

Manual transformation: The use of coding languages to affect data transformation, without the aid of software programs

Master data management (MDM): A discipline and framework for managing and maintaining critical data assets in order to achieve a single, consistent view of data across the organization

Measured service: A service that charges users only for what they use, based on the number of transactions, the storage volume, and the amount of data transferred

Measures: Aggregations of one or more dimensions, like a count or average

Metadata: Data about data

Metadata management: The process of managing, accessing, and organizing metadata

Missing data: When the rows you expect to be returned do not appear in the joined table

Modularity: A concept that breaks down a system into smaller parts that can be easily separated and reused

N

Nested data structure: A structure that organizes data within other data structures, forming a hierarchy of information

Network use: The amount of data that is read or moved between storage buckets

Normalized data: Organizes related fields into different tables, and maintains defined relationships between columns in these different tables

NULL value: No available value for a field

Numerical data (Quantitative data): A specific and objective measure, like a number, quantity, or range

O

On-premises: Information technology infrastructure that is physically located in an organization's own data center or office

Organization-wide metrics: Metrics that are defined and shared across teams

Outer join: A join that returns both matched and unmatched rows from one or both tables

Output data: Any information that a computer sends to a user

P

Pandas: An open source data analysis and manipulation tool that can be used with Python

Parent-child relationship: A feature of a ticketing system that enables users to divide a single ticket into subtickets, which can be worked on simultaneously

Partial duplicate: A record where only part of the data repeats another record

Partition pruning: The process of eliminating unnecessary or irrelevant data from consideration when running a query

Partitioned table: A large table that is divided into smaller segments, or partitions, to improve query performance, and control costs by reducing the amount of data a query needs to process

Personally identifiable information (PII): Data that is unique to an individual, and therefore can be used to distinguish one person from another

Platform as a service (PaaS): A cloud computing model that provides hardware and software tools to create an environment for the development of cloud applications, simplifying the application-development process

Primary key: An identifier that references a column in which each value is unique

Private cloud: A cloud model that dedicates all cloud resources to a single user or organization, and is created, managed, and owned within on-premises data centers

Protected health information (PHI): Health data that can identify an individual, like information about patient demographics, mental or physical health diagnoses or treatments, and payment records related to health care

Public cloud: A cloud model that delivers computing, storage, and network resources through the internet

PySpark: A language for creating more automated analyses and pipelines

Python: A high-level, general-purpose programming language

R

R: A programming language used for statistical computing

Ranking: The process of using characteristics to order items by likelihood

Refactoring: A cloud-migration strategy that involves building all-new applications from scratch and discarding old applications

Rehosting or "lift and shift": A cloud-migration strategy that involves moving an entire on-premises system to the cloud

Relational database: A database that contains a series of tables that can be connected to form relationships

Replatforming: A cloud-migration strategy that involves making small changes to the on-premises system once it's migrated to the cloud

Repository: A central location for storing and managing the files and history of a project

Repurchasing: A cloud-migration strategy that involves moving applications to a new, cloud-based service platform, usually a software-as-service

Reserved instances: A cloud payment model in which an organization purchases a specific amount of resources for a certain time period, and receives a discount in return for this commitment

Resource provisioning: The process of a user selecting appropriate software and hardware resources, and the cloud service provider setting them up and managing them while in use

Retention policy: An outline of how data is archived and deleted

Retiring: Applications that are no longer useful are turned off

Rightsizing: The process of adjusting computing resources to fit the exact needs of an application or workload

S

Sampling: The technique of selecting a segment of a dataset that is representative of the entire dataset in order to better understand its characteristics

Schema: A way of describing how something, like data, is organized

Scorecard: A statistical record used to measure achievement or progress toward a goal

Security key: An authentication method that uses a physical digital signature, or key, to verify a user's identity before accessing specific resources

Self-provisioning: A cloud delivery model in which the user purchases resources from the cloud provider through a website or online portal, then the resources are quickly made available for the user

Self-service analytics: An approach to business intelligence that allows both technical and non-technical users across an organization to access data, perform ad-hoc data analysis, and generate reports

Semantic data model: A type of data model that uses everyday language to represent data

Semantic layer: A set of definitions and logic that helps ensure that everyone in the organization understands the data in the same way

Semi-structured data: Data that has some structure but is not as precisely organized as structured data

Software as a service (SaaS): A cloud computing model that provides users with a licensed subscription to a complete software package

Software development kit (SDK): A set of software-building tools that includes groups of code libraries

SQL dialect: A version of SQL that is unique to a specific database

Streaming data processing: A method of processing data as it's received

Streaming ingestion: A method of collecting and processing data as soon as it becomes available

Streaming loading: A method where the data is moved to destination storage in a continuous stream, as soon as it becomes available

String: A sequence of characters and punctuation that contains textual information

Struct: A collection of named fields

Structured data: Data organized in a certain format, like rows and columns

Structured Query Language (SQL): Used to get data out of relational databases

Syntax: The predetermined structure of a language that includes all required words, symbols, and punctuation, and their proper placement

System of record: A data-storage system that serves as the source of truth for an organization's data related to processes or systems

T

Time sensitive data: Data that must be acted on within a specific time frame, or it loses value

Traditional computing: A computing model that enables data storage, access, and management through the use of physical hardware and software within a network infrastructure, typically located on-premises

Transactional database: A data-storage system that maintains individual rows of data, rather than columns

Transform: The stage of taking data, cleaning it, and putting it into a standard format in a data pipeline

U

Unstructured data: Data that is not organized in any easily identifiable way

Uptime: The amount of time a machine is operational

User experience (UX): The overall experience a user has with a product or service, especially in terms of how easy or pleasing it is to use

User interface (UI): The means through which a user and a computer system interact

V

Validation rules: A set of instructions that specify the standards that must be met for data to be valid, and how to handle data errors

Version control: A process to track changes to your code, data, or other files over time

Versioning: The process of creating a unique way of referring to data

Vertical partitioning: A process for dividing data into separate segments, in which each partition maintains a unique piece of the fields for items in the data store

Virtualization: Technology that creates a virtual version of physical infrastructure, like servers, storage, and networks

W

Wireframe: A visual representation of the structure and functionality of a user interface or product layout